

06/08/2025



Code-C Phase-3

Aakash

Medical | IIT-JEE | Foundations

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MM : 720

Fortnightly Test for NEET-2026_RM(P3)_FT-01C

Time : 180 Min.

Topics Covered:

Physics: Units & Measurements, Motion in a Straight Line

Chemistry: Some Basic Concepts of Chemistry

Botany: Cell: The Unit of Life

Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules-I: (Upto polysaccharides)

General Instructions :

Duration of Test is 3 hrs.

The Test consists of 180 questions. The maximum marks are 720.

There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.

Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.

Use blue/black ballpoint pen only to darken the appropriate circle.

Mark should be dark and completely fill the circle.

Dark only one circle for each entry.

Dark the circle in the space provided only.

Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. Which of the following relation is correct?

(1) $1 \text{ ly} = 1.5 \times 10^{11} \text{ m}$

(2) $1 \text{ AU} = 9.46 \times 10^{15} \text{ m}$

(3) $1 \text{ parsec} = 3.26 \text{ ly}$

(4) $1 \text{ ly} = 3.1 \times 10^{16} \text{ m}$

2. In a Vernier calliper 20 Vernier scale divisions are equal to 19 main scale divisions and each main scale division is of 1 mm. The least count of the Vernier calliper is

(1) 0.05 mm

(2) 1 mm

(3) 0.5 mm

(4) 2 mm

3. A micrometer has 150 equal divisions on the circular scale, and on one full rotation of circular scale main scale advances by 0.02 cm on the main scale. The least count of the micrometer (in cm) is

(1) 4×10^{-5}

(2) 6.21×10^{-3}

(3) 1.33×10^{-4}

(4) 1.21×10^{-2}

4. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above reference level. The correct diameter of ball is

(1) 0.521 cm

(2) 0.525 cm

(3) 0.529 cm

(4) 0.053 cm

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5. Which of the following statement(s) is/are correct?

Statement I: A vernier calliper is more precise than a meter scale.

Statement II: Lower value of least count of an instrument result in more error.

- (1) Only (I)
 (2) Only (II)
 (3) Both (I) and (II)
 (4) Neither (I) nor (II)
6. In a new system, unit of length is 6 m, unit of mass is 2 kg, unit of time is 2 minute. In the new system of units, unit of pressure is equal to
- (1) $12 \text{ kgm}^{-1} \text{ min}^{-2}$
 (2) $24 \text{ kgm}^{-1} \text{ min}^{-2}$
 (3) $\frac{1}{24} \text{ kgm}^{-1} \text{ min}^{-2}$
 (4) $\frac{1}{12} \text{ kgm}^{-1} \text{ min}^{-2}$

7. In column-I few measurements are given and in column-II, the number of significant figures are given. Match the column-I and column-II.

	Column-I		Column-II
a.	0.0007 m^2	(i)	3
b.	$2.64 \times 10^{24} \text{ kg}$	(ii)	5
c.	6.320 J	(iii)	1
d.	$0.63200 \text{ g cm}^{-3}$	(iv)	4

- (1) a(i), b(ii), c(iv), d(iii)
 (2) a(iii), b(i), c(iv), d(ii)
 (3) a(iii), b(iv), c(i), d(ii)
 (4) a(iv), b(ii), c(iii), d(i)
8. Among the following, which is not a fundamental quantity?
- (1) Length
 (2) Luminous intensity
 (3) Electric current
 (4) Charge
9. A wire has a mass of $(0.3 \pm 0.003) \text{ g}$, radius $(0.5 \pm 0.005) \text{ mm}$ and length $(6 \pm 0.06) \text{ cm}$, then the percentage error in the density of the wire is given by
- (1) 2%
 (2) 5%
 (3) 4%
 (4) 10%

10. If percentage error in radius of a sphere is 0.2%, then the percentage error in its volume is

- (1) 0.2%
 (2) 0.4%
 (3) 0.6%
 (4) 0.8%

11. A physical quantity $P = \frac{\sqrt{ABC^2}}{D^2}$ is determined by measuring A, B, C and D separately with the percentage error of 2%, 2%, 4% and 1% respectively. Maximum percentage error in the value of P will be

- (1) 7%
 (2) 9%
 (3) 12%
 (4) 6%

12. The radius of a circle varies with time t is given as $r = t^2 + 2t + 1$. The rate of change of area of the circle at $t = 1 \text{ s}$ will be (where r is in meter and t is in second)

- (1) $32\pi \text{ m}^2/\text{s}$
 (2) $16\pi \text{ m}^2/\text{s}$
 (3) $24\pi \text{ m}^2/\text{s}$
 (4) $8\pi \text{ m}^2/\text{s}$

13. In the addition of three masses 1.4 g, 4.24 g and 7.432 g has x significant figures. The value of x is

- (1) 2
 (2) 3
 (3) 4
 (4) 5

14. The value of $(2.5 \times 10^{-3} - 4.40 \times 10^{-4})$, with regard to significant figure is

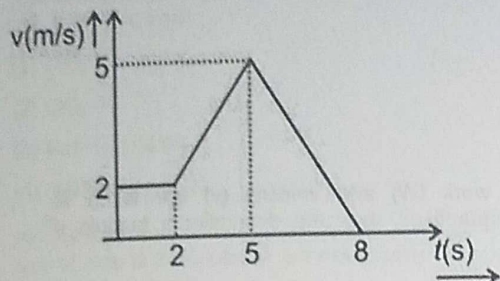
- (1) 2.06×10^{-3}
 (2) 2.0×10^{-3}
 (3) 2.1×10^{-3}
 (4) 2.1×10^{-4}

15. In an experiment measurements of velocity of an object (in ms^{-1}) are 342, 338, 318 and 322. The mean absolute error in the measurement is

- (1) 8 ms^{-1}
 (2) 4 ms^{-1}
 (3) 10 ms^{-1}
 (4) 12 ms^{-1}

16. The most accurate reading of mass 25.0 g is
 (1) 24.00 g
 (2) 24.800 g
 (3) 25.1 g
 (4) 25.20 g
17. The value of $\frac{36.3 \times 12}{4.34}$ upto correct significant figure is
 (1) 1.0×10^2
 (2) 1.00
 (3) 100.4
 (4) 100.0
18. Time period of simple pendulum is given by $T = 2\pi \sqrt{\frac{l}{g}}$, where l is the length of pendulum and g is acceleration due to gravity. If percentage error in the length and acceleration due to gravity is 4% and 2% respectively, then the percentage error in time period is
 (1) 3%
 (2) 4%
 (3) 6%
 (4) 1%
19. If C and L denote the capacity and inductance, then the dimension of LC are
 (1) $M^0 L^0 T^2$
 (2) $ML^2 T^{-2}$
 (3) $M^2 L^2 T^{-2}$
 (4) MLT^{-1}
20. In relation $y(x, t) = A \sin(\omega t - kx)$, dimension of (ω/k) will be equal to (Here x is position and t is time)
 (1) $[M^0 L^0 T^0]$
 (2) $[M^0 L^1 T^{-1}]$
 (3) $[M^0 L^{-1} T^{-1}]$
 (4) $[M^0 L^0 T^1]$
21. Kinetic energy of an object moving along a straight line is given by $K = \frac{A\sqrt{x}}{x^2 + B}$, where A and B are dimensional constants and x is position of object. The dimensional formula for $\frac{A}{B}$ is
 (1) $[M^1 L^{7/2} T^{-2}]$
 (2) $[M^{-1} L^{-3/2} T]$
 (3) $[M^{-1} L^{-7/2} T^2]$
 (4) $[ML^{3/2} T^{-2}]$
22. If the unit of force is 10 kN, length is 10 km and time is 1000 s, then the unit of mass will be
 (1) 10^3 kg
 (2) 10^4 kg
 (3) 10^5 kg
 (4) 10^6 kg
23. If force (F), work (W) and velocity (v) are taken as fundamental quantities, then the dimensional formula of time (T) is
 (1) $[W F v]$
 (2) $[W F^{-1} v^{-1}]$
 (3) $[W^{-1} F v^{-2}]$
 (4) $[W F^2 v^{-1}]$
24. The velocity of a particle depends upon the time t according to the equation $v = \sqrt{at} + bt + \frac{c}{d+t}$. The physical quantities which are represented by a, b, c , and d are in the following order
 (1) Distance, distance, acceleration, time
 (2) Distance, acceleration, distance, time
 (3) Acceleration, distance, distance, time
 (4) Time, distance, acceleration, distance
25. The distance travelled by a particle is proportional to the square of time then the particle travels with
 (1) Uniform velocity
 (2) Uniform acceleration
 (3) Increasing acceleration
 (4) Decreasing acceleration
26. The graph between velocity and time, if the position of the particle varies as $x = 4t^2$, will be
 (1) Hyperbola
 (2) Ellipse
 (3) Parabola
 (4) Straight line
27. A : The speed of particle is increasing if acceleration is positive.
 R : The speed of particle is decreasing if acceleration is negative.
 (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

28. Velocity-time ($v-t$) graph for a moving object is shown in the figure. The net displacement of the object, during the time interval when there is non zero acceleration, will be



- (1) 9 m
(2) 18 m
(3) 6 m
(4) 12 m
29. A motor vehicle left the point A and reached the point B by travelling in a straight line for 2 hours. The vehicle travelled half of the distance at a speed of $v_1 = 40$ km/h and other half at a speed of $v_2 = 60$ km/h. The distance between A and B is

- (1) 86 km
(2) 96 km
(3) 76 km
(4) 106 km

30. The velocity of a particle is given by $V = \sqrt{150 - 20x}$, where 'x' is position in m and V in m/s. Its acceleration is

- (1) -8 m/s^2
(2) -10 m/s^2
(3) 12 m/s^2
(4) -12 m/s^2

31. The position x of a particle varies with time t as $x = at^2 - bt^3$, where a and b are constants. The acceleration of the particle will be equal to zero, when t is equal to (All quantities are in SI units)

- (1) $\frac{a}{2b}$
(2) $\frac{a}{3b}$
(3) $\frac{a}{6b}$
(4) $\frac{a}{b}$

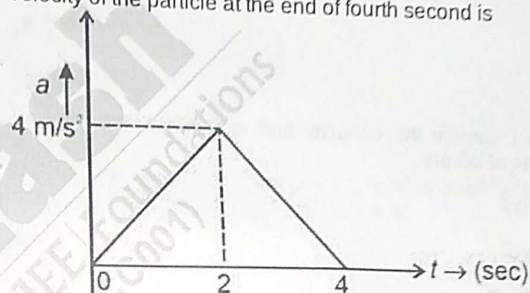
32. A car moving with speed 30 m s^{-1} can be stopped by the application of brakes in a distance of 81 m. If the speed of the car is 20 m s^{-1} , it can be stopped in (Assuming uniform retardation)

- (1) 36 m
(2) 81 m
(3) 72 m
(4) 60 m

33. A body A starts from rest with acceleration a_1 . After two second another body B starts from rest with acceleration a_2 . If they travel equal distance in the 5th second after the start of A. Then the ratio of $a_1 : a_2$ is equal to

- (1) 5 : 9
(2) 1 : 3
(3) 3 : 1
(4) 5 : 1

34. Acceleration-time graph of a particle moving in a straight line is shown below. Velocity of particle at $t = 0$ is 5 m/s. The velocity of the particle at the end of fourth second is

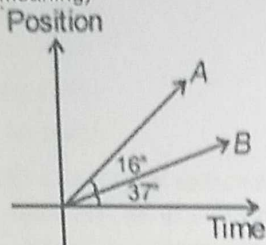


- (1) 16 m/s
(2) 11 m/s
(3) 13 m/s
(4) 5 m/s

35. The numerical ratio of average speed to average velocity of an object is

- (1) Equal to one
(2) More than one
(3) Less than one
(4) More than or equal to one

36. The position-time graph for two particles A and B are as follows. The ratio of $\frac{v_A}{v_B}$ is (All symbols have their usual meaning)



- (1) 1
(2) $\frac{4}{3}$
(3) $\frac{16}{9}$
(4) $\frac{3}{4}$
37. A car travels first 10 seconds with uniform acceleration of 2 m/s^2 from rest on straight road and then travels with zero acceleration for next 10 seconds. The average speed of entire journey is
- (1) 20 m/s
(2) 15 m/s
(3) 10 m/s
(4) 5 m/s
38. A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and eventually comes to rest. The correct value of $\frac{a_1}{a_2}$ will be
- (1) $\frac{t_1+t_2}{t_1}$
(2) $\frac{t_1}{t_2}$
(3) $\frac{t_2}{t_1}$
(4) $\frac{t_1+t_2}{t_2}$
39. A body travels distance of 20 m and 40 m with constant speed of 5 m s^{-1} and 10 m s^{-1} respectively, the average speed of the body is
- (1) 6.5 m s^{-1}
(2) 3.5 m s^{-1}
(3) 7.5 m s^{-1}
(4) 9 m s^{-1}

40. If an object is released from rest from a height, then ratio of distance travelled by it in 1st second, 2nd second and 3rd second of motion will be

- (1) 1 : 2 : 3
(2) $1^2 : 2^2 : 3^2$
(3) 1 : 3 : 5
(4) 1 : 1 : 1

41. A car starts from rest and accelerates at 1 ms^{-2} for 4 s. It then moves at a constant velocity for next 10 s, then decelerates at 2.0 ms^{-2} and finally comes to rest. The total distance travelled by the car is

- (1) 52 m
(2) 40 m
(3) 48 m
(4) 46 m

42. A juggler maintains 10 balls in motion, making each of them to rise at a height of 80 m from his hands. To keep proper distance between them, the time interval maintained by juggler would be

- (1) 0.6 s
(2) 0.8 s
(3) 1.0 s
(4) 1.2 s

43. A bus starts from rest with an acceleration of 1 ms^{-2} . A boy who is 48 m behind the bus, starts with uniform speed of 10 m/s. The time after which he will catch the bus is

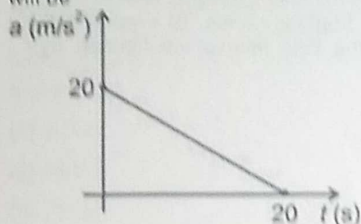
- (1) 4 s
(2) 6 s
(3) 7 s
(4) 8 s

44. Match list I with list II and tick the correct option. Here x , v and t represent position, velocity and time respectively.

	List I		List II
a.	Acceleration	(i)	$\left \frac{dx}{dt} \right ^3$
b.	Square of speed	(ii)	$\frac{dx}{dt}$
c.	Velocity	(iii)	$\frac{d^2x}{dt^2}$
d.	Rate of change in speed	(iv)	$\frac{d v }{dt}$

- (1) a(iii), b(i), c(ii), d(iv)
 (2) a(ii), b(i), c(iii), d(iv)
 (3) a(i), b(ii), c(iii), d(iv)
 (4) a(iv), b(i), c(ii), d(iii)

45. A particle starts from rest. Its acceleration (a) versus time (t) curve is shown in the figure. Maximum speed of the particle will be



- (1) 400 m/s
 (2) 100 m/s
 (3) 200 m/s
 (4) 500 m/s

CHEMISTRY

46. Which has the maximum number of molecules among the following?

- (1) 8 g H_2
 (2) 64 g SO_2
 (3) 44 g CO_2
 (4) 48 g O_3

47. Two compounds AB_3 and A_2B are formed by the combination of two elements A and B. If 0.1 mol AB_3 weighs 11 g while 0.5 mol of A_2B weighs 35 g then the atomic weights of A and B respectively are

- (1) 25u and 30u
 (2) 20u and 30u
 (3) 25u and 40u
 (4) 20u and 40u

48. A water bath has temperature $65^\circ C$. Its value in $^\circ F$ scale will be

- (1) $142^\circ F$
 (2) $165^\circ F$
 (3) $149^\circ F$
 (4) $155^\circ F$

49. What will be the empirical formula of the compound formed by 'X' and 'Y' if it has 32% of X?
 (Given that atomic mass of X = 55 u, Y = 40 u)

- (1) X_3Y_2
 (2) XY_3
 (3) X_2Y_3
 (4) X_3Y

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50. Volume of O_2 required at STP to react completely with 560 g of ethene (C_2H_4) is
- 2245 L
 - 448 L
 - 1344 L
 - 1860 L
51. What is mass of sodium acetate (CH_3COONa) required to make 200 ml of 0.245 molar aqueous solution? (Molar mass of $CH_3COONa = 82 \text{ g mol}^{-1}$)
- 3 g
 - 4 g
 - 5 g
 - 6 g
52. Assertion (A) : Empirical formula shows the exact number of different types of atoms present in a molecule of a compound.
Reason (R) : Molecular formula represents the simplest whole number ratio of various atoms present in a compound.
In the light of above statements, choose the correct answer.
- Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 - Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 - Assertion is true statement but Reason is false
 - Both Assertion and Reason are false statements
53. How many moles of aluminium sulphate, $Al_2(SO_4)_3$ will contain 0.24 mole of oxygen atoms?
- 2.0×10^2
 - 2.0×10^{-2}
 - 3.0×10^3
 - 1.5×10^{-2}
54. Which among the following concentration terms is temperature dependent?
- Mole fraction
 - Mass per cent
 - Molality
 - Molarity
55. The number of significant figures in 2.30×10^{-15} , 230 and 0.230 respectively are
- 3, 3 and 3
 - 3, 2 and 3
 - 2, 3 and 4
 - 2, 2 and 4
56. If the mole fraction of acetone in ethanol is 0.2 then molality of acetone in ethanol will be
- 5.4 m
 - 6.8 m
 - 4.5 m
 - 7.2 m
57. From 200 mg of CO_2 when x molecules are removed, 2.89×10^{-3} moles of CO_2 are left. x will be
- 10^{20} molecules
 - 10^{10} molecules
 - 21 molecules
 - 10^{21} molecules
58. Element X has two isotopes A and B. If % abundance of A and B is 50% each, then average atomic mass of element X is (At. mass of A = 10 u and B = 20 u)
- 10 u
 - 15 u
 - 20 u
 - 30 u
59. The mass percentage of sulphur in sulphuric acid is
- 32.65%
 - 54.32%
 - 15.68%
 - 39.75%
60. The correct scientific notation for 231.508 is
- 2.31508×10^2
 - 2.31508×10^{-2}
 - 0.231508×10^3
 - 0.231508×10^{-3}
61. Number of valence electrons present in 16 mg of oxygen atoms is
- 6.02×10^{21}
 - 6.02×10^{20}
 - 3.6×10^{21}
 - 3.6×10^{20}

62. Given below are two statements labelled as Assertion (A) and Reason (R).

Assertion (A): One mole of methane contains 4 moles of atoms of methane.

Reason (R): One mole is amount of substance that contains as many constituent particles as number of atoms present in 12 g of ^{12}C isotope

Select the most appropriate answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A
 - (2) Both A and R are true but R is not the correct explanation of A
 - (3) A is true but R is false
 - (4) A is false but R is true
63. If the equivalent weight of a metal hydride is 13, then the equivalent weight of metal oxide is
- (1) 10
 - (2) 18
 - (3) 19
 - (4) 20

64. Which of the following is the incorrect match?

	Prefix	Multiple
(1)	Milli	10^{-3}
(2)	Pico	10^{-12}
(3)	Nano	10^{-9}
(4)	Micro	10^{-4}

- (1) (1)
 - (2) (2)
 - (3) (3)
 - (4) (4)
65. 100 ml of 20.8% (w/v) BaCl_2 (aq) and 100 mL of 14.7% (w/v) H_2SO_4 (aq) solutions are mixed. Molarity of SO_4^{2-} ions in the resulting solution is [Molar mass of $\text{BaCl}_2 = 208 \text{ g/mol}$]
- (1) 0.5 M
 - (2) 0.1 M
 - (3) Zero
 - (4) 0.25 M
66. If the mass ratio of H_2 and CH_4 in a mixture is 1 : 4, then their mole ratio will be
- (1) 1 : 4
 - (2) 2 : 1
 - (3) 1 : 2
 - (4) 1 : 1

67. 3 g of an oxide of metal is converted completely into its chloride having mass 5 g. Equivalent weight of metal will be

- (1) 8
 - (2) 40
 - (3) 35.55
 - (4) 33.25
68. What is the molarity of H_2SO_4 solution containing 49% H_2SO_4 by mass? (density of solutions = 1.09 g/mL)
- (1) 20 M
 - (2) 5.45 M
 - (3) 15 M
 - (4) 13.80 M
69. 4 g of Mg is reacted with 8 g of O_2 . Mole of MgO produced in the reaction is

- (1) $\frac{1}{4}$
 - (2) $\frac{1}{8}$
 - (3) $\frac{1}{6}$
 - (4) $\frac{1}{3}$
70. 5 g impure NaOH is completely neutralised by one litre 0.1 M HCl. The percentage purity of NaOH sample is
- (1) 80%
 - (2) 60%
 - (3) 40%
 - (4) 20%

71. Select the **incorrect** statement regarding Dalton's atomic theory

- (1) Matter consists of indivisible atoms
- (2) All atoms of a given element have identical properties
- (3) Compounds are formed when atoms of different elements combine in fixed ratio
- (4) Atoms are created or destroyed in chemical reactions

72. Which of the following is a unitless quantity?

- (1) Normality
- (2) Molality
- (3) Molarity
- (4) Mole fraction

73. Heterogeneous mixture among the following is/are

- (1) Mixture of $\text{O}_2(\text{g})$ and $\text{N}_2(\text{g})$
- (2) Sugar solution
- (3) Brass
- (4) Mixture of grains and pulses

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74. Law of Multiple proportion is applicable to which pair of compounds?

- (1) H_2S and H_2O
- (2) HCl and HBr
- (3) SO_2 and SO_3
- (4) NaCl and KCl

75. Match the following

	Column I		Column II
a.	Luminous intensity	(i)	Kelvin
b.	Frequency	(ii)	Candela
c.	Temperature	(iii)	Ampere
d.	Electric current	(iv)	Mole
e.	Amount of substance	(v)	Hertz

Choose the correct option

- (1) a(iii), b(v), c(i), d(ii), e(iv)
- (2) a(ii), b(v), c(i), d(iii), e(iv)
- (3) a(v), b(iv), c(ii), d(iii), e(i)
- (4) a(v), b(iv), c(i), d(iii), e(ii)

76. Given below are two statements

Statement I: Both 12 g of carbon and 27 g of Al will have 6.02×10^{23} atoms.

Statement II: Gram atomic mass of an element contains Avogadro's number of atoms.

In the light of above statements choose the correct option.

- (1) Both Statement I and statement II are correct.
- (2) Both statement I and statement II are incorrect.
- (3) Statement I is correct but statement II is incorrect.
- (4) Statement I is incorrect but statement II is correct.

77. Select the one which is not a compound?

- (1) Water
- (2) Mercury
- (3) Methane
- (4) Ammonia

78. Consider the following statements

Statement-I: Gay Lussac's Law of Gaseous Volumes : When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided that all gases are at same temperature and pressure.

Statement-II: The volumes of $\text{H}_2(\text{g})$ and $\text{O}_2(\text{g})$ which combine together to form water bear a simple ratio of 2 : 1 In the light of above statements, choose the correct answer from the options given below.

- (1) Statement I is false but statement II is true
- (2) Both statements I and II are true
- (3) Both statements I and II are false
- (4) Statement I is true but statement II is false

79. For which of the following compounds empirical formula and molecular formula are same?

- (1) CH_3COOH
- (2) $\text{HOCH}_2\text{CH}_2\text{OH}$
- (3) C_6H_6
- (4) CH_3CHO

80. 100 mL of 2 M H_2SO_4 solution is diluted by adding 400 ml water. The molarity of the resultant solution becomes

- (1) 0.4 M
- (2) 0.6 M
- (3) 0.75 M
- (4) 0.2 M

81. The crystalline salt $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ on heating loses 63% of its mass and becomes anhydrous. The formula of the compound is

- (1) $\text{Na}_2\text{CO}_3 \cdot 2\text{H}_2\text{O}$
- (2) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- (3) $\text{Na}_2\text{CO}_3 \cdot 5\text{H}_2\text{O}$
- (4) $\text{Na}_2\text{CO}_3 \cdot 8\text{H}_2\text{O}$

82. A chemist wants to prepare 0.35 M HCl . Commercial HCl is 15.5 M. How many millilitres of the commercial acid does the chemist require to make up 1.8 L of the dilute acid of 0.35 molarity?

- (1) 4.06 ml
- (2) 40.6 ml
- (3) 112.3 ml
- (4) 1.12 ml

83. Maximum mass among the following is of

- (1) 12 g-atoms of O
- (2) 6 g-molecules of O_2
- (3) 5 mole of O_3
- (4) 224 L of O_2 at STP

84. Mass of four atoms of helium is

- (1) 4 Na
- (2) 4 u
- (3) 16 u
- (4) 32 u

85. Number of Fe atoms in 22.4 g Haemoglobin if it contains 0.25% Fe by mass (Atomic mass of Fe = 56 u) is

(1) 6.02×10^{21}
 (2) 6.02×10^{20}
 (3) 3.01×10^{20}
 (4) 3.01×10^{21}

86. If mass percentage of zinc in a biomolecule is 0.4% then the minimum molecular mass of the biomolecule will be (Atomic mass of zinc = 65.4 u)

(1) 12250 u
 (2) 22275 u
 (3) 16350 u
 (4) 6250 u

87. Consider the following statements:

Statement I: 100 g of calcium carbonate (CaCO_3) on heating gives 44 g of carbon dioxide and 56 g of calcium oxide (CaO).

Statement II: During a chemical change, total mass of product obtained is equal to the total mass of the reactant(s) taking part in the reaction.

In the light of above statements, choose the correct option.

- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

88. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R)
 A : In a chemical reaction, limiting reagent decides the yield of the product.

R : Limiting reagent is partially consumed in a reaction.

In the light of above statements, choose the correct answer.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
 (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
 (3) Assertion is true statement but Reason is false
 (4) Both Assertion and Reason are false statements

89. At STP 11.2 L of an unknown gas weighs 15 g. Unknown gas may be

(1) NO_2
 (2) C_2H_6
 (3) CO_2
 (4) C_2H_2

90. How many g of dibasic acid (mol. weight 200) should be present in 100 ml. of the aqueous solution to give strength of 0.1 N?

(1) 10 g
 (2) 2 g
 (3) 1 g
 (4) 20 g

BOTANY

91. Match the following columns and select the correct option

Column I

- a. Metacentric chromosome
 b. Acrocentric chromosome
 c. Sub-metacentric chromosome
 d. Telocentric chromosome

Column II

- (i) Centromere is situated close to its one end
 (ii) Centromere is slightly away from the middle
 (iii) Appears to have a single arm
 (iv) Has two equal arms

- (1) a(iv), b(ii), c(i), d(iii)
 (2) a(iv), b(i), c(ii), d(iii)
 (3) a(ii), b(i), c(iv), d(iii)
 (4) a(ii), b(iv), c(i), d(iii)

92. Which of the following statement is incorrect?

- (1) Inner membrane of mitochondria forms a number of foldings called cristae
 (2) Middle lamella of plants is composed of chitin
 (3) Plasmodesmata connect the neighbouring plant cells
 (4) Mitochondria are the sites of aerobic respiration

93. The term glycocalyx is used for

- (1) A layer surrounding plant cell.
 (2) Cell wall of bacteria.
 (3) A layer surrounding bacterial cell wall.
 (4) Outermost layer of animal cell

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94. Which one is **odd** w.r.t. prokaryotic flagella?

- (1) Tubulin protein
- (2) Basal body consists of rings
- (3) Hook
- (4) Help in motility

95. Nucleolus is an active site for the synthesis of

- (1) Ribosomes
- (2) Ribosomal RNA
- (3) Double stranded DNA
- (4) Centriole

96. Which of the following statements is **incorrect** w.r.t. the outermost layer of prokaryotic cell envelope?

- (1) The layer differs in thickness and chemical composition in different bacteria
- (2) It protects the cell from loss of water and nutrients
- (3) It protects the host's immune system from adverse effects of bacteria
- (4) It protects the bacterial cell and also helps in adhesion

97. Which one of the following is **not** a function of cell wall in plant cells?

- (1) Protects the cell from infection
- (2) Helps in cell-to-cell interaction
- (3) Controls molecular transport within the cell
- (4) Provides barrier to undesirable macromolecules

98. Consider the following statements and choose the **correct** option.

Statement A: The interphase nucleus has a loose and extended network of nucleo-protein fibres called chromatin.
Statement B: Sometimes a few chromosomes have non-staining secondary constriction at a constant location.

- (1) Only statement A is incorrect
- (2) Only statement B is incorrect
- (3) Both the statements are correct
- (4) Both the statements are incorrect

99. Match List-I with List-II.

	List I		List II
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the **correct** answer from the options given below.

- (1) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)
- (2) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)
- (3) (a)–(i), (b)–(iv), (c)–(iii), (d)–(ii)
- (4) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

100. Which type of substance would face difficulty to pass through the cell membranes?

- (1) Substance soluble in lipids
- (2) Substance with hydrophobic moiety
- (3) Substance with hydrophilic moiety
- (4) All substance irrespective of hydrophobic and hydrophilic moiety

101. Main arena of cellular activities in both prokaryotes and eukaryotes is

- (1) Plasma membrane
- (2) Cytoplasm
- (3) Nucleus
- (4) Cell wall

102. Pumping out of sodium ions through the cell membrane

- (1) Is a passive transport
- (2) Does not require the ATP energy
- (3) Occurs by a process of simple diffusion
- (4) Is an energy utilizing process

103. The nuclear pores are complex structure which are formed by the

- (1) Disintegration of nuclear membrane
- (2) Fusion of outer and inner membranes of nucleus
- (3) Enzymatic action of lysosome
- (4) Movement of RNA and protein molecules

104. Which of the following is **incorrect** about the organelle that gives rise to spindle apparatus during cell division?

- (1) It is surrounded by amorphous pericentriolar materials
- (2) It usually contains two cylindrical structures, that lie parallel to each other
- (3) Cartwheel like organisation can be observed
- (4) Each peripheral fibril is a triplet, connected to proteinaceous hub

105. Select the **correctly** matched pair.

- (1) Microtubule – Formation of furrow during cell division
- (2) Microfilament – Form cytoskeleton of cilia
- (3) Intermediate filament – Pseudopodia formation
- (4) Microtubule – Determining cell plate position

106. Which of the following statements was given by Rudolf Virchow to modify the cell theory in its final shape?

- (1) All the living cells have nucleic acid as genetic material.
- (2) Cells divide and new cells are formed from pre-existing cells.
- (3) All cells are covered by an envelope.
- (4) All living organisms are composed of cells and products of cells.

107. Thick and tough glycocalyx found in some bacteria is called

- (1) Slime layer
- (2) Mesosome
- (3) Capsule
- (4) Lamellae

108. Which among the following is the function of the cytoskeleton?

- (1) Protein synthesis
- (2) Motility
- (3) Lipid synthesis
- (4) mRNA synthesis

109. Read the following statements.

Statement A: The fluid nature of the plasma membrane helps in the formation of the intercellular junction, secretion, endocytosis and cell division.

Statement B: In plasma membrane, arrangement of lipids ensures that the non-polar tails of saturated hydrocarbons are protected from the aqueous environment.

In the light of above statements, choose the **correct** option.

- (1) Statement A is true but statement B is false
- (2) Statement B is true but statement A is false
- (3) Both statement A and statement B are true
- (4) Both statement A and statement B are false

110. Select the **correct** statement(s) regarding mitochondria.

- (a) Mitochondria show a great degree of variability in their shape and size but their number remains constant per cell.
- (b) Outer membrane is composed of 80% proteins and 20% lipids.
- (c) The enzymes and electron carriers for the formation of ATP are present only in the inner membrane.
- (d) These are miniature biochemical factories where proteins and fats are completely reduced.

(1) (a), (b) and (c)

(2) (b), (c) and (d)

(3) (c) and (d) only

(4) (c) only

111. Chromatin contains

- (1) DNA and basic proteins only
- (2) DNA and RNA only
- (3) DNA, RNA, histone and non-histone proteins
- (4) RNA and histone proteins only

112. All of the following are inclusion bodies in prokaryotic cells, **except**

- (1) Phosphate granules
- (2) Cyanophycean granules
- (3) Glycogen granules
- (4) Sap vacuoles

113. Which of the given descriptions is **not** related to the organelles of endomembrane system?

- (a) Organelle occupying upto 90% of the volume of plant cell.
- (b) Organelle called as palade particles
- (c) Organelle with peroxide producing as well as destroying enzymes.
- (d) Organelle named as dictyosomes in plants.

(1) (b) & (d)

(2) (b) & (c)

(3) (a) & (d)

(4) (a) & (c)

114. Membranous extensions that contain photosynthetic pigments in some prokaryotes are

- (1) Mesosomes
- (2) Chromatophores
- (3) Granules
- (4) Filament

115. Read the following Assertion (A) and Reason (R) and select the correct option.

Assertion (A): Concentration of ions and molecules is significantly higher in the vacuole than in the cytoplasm.

Reason (R): Tonoplast membrane facilitates the transport of a number of ions and other materials against concentration gradients into the vacuole.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

116. Match the structures given in column-I with their functions/explanations given in column-II, and select the most appropriate answer.

Column I

Column II

- | | |
|---------------------|---|
| a. Inclusion bodies | (i) Increases surface area and enzymatic content of plasma membrane |
| b. Plasmid DNA | (ii) Several ribosomes attached to a single mRNA |
| c. Polysome | (iii) Reserve materials in the cytoplasm of prokaryotes |
| d. Mesosome | (iv) Confers certain unique phenotypic characters to bacteria |

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(iii), b(iv), c(ii), d(i)
- (4) a(iii), b(ii), c(iv), d(i)

117. Which of the following has nucleus?

- (1) *Mycoplasma*
- (2) Yeast
- (3) Blue green algae
- (4) PPLO

118. Peroxisomes are not considered as parts of the endomembrane system because

- (1) It is not involved in carbohydrate oxidation
- (2) Its functions are not coordinated with the functions of cell organelles of endomembrane system
- (3) They do not perform photosynthesis
- (4) It is not surrounded by double membrane

119. The size of a typical eukaryotic cell is

- (1) 1-2 μm
- (2) About 3.0 μm
- (3) 0.02 – 0.2 μm
- (4) 10-20 μm

120. State True (T) or False (F) to the given statements and select the correct option.

- (A) Anton von Leeuwenhoek was the first person who discovered living cells
- (B) Matthias Schleiden, studied different types of animal cells and reported that cells had thin outer layer known as plasma membrane
- (C) The cytoplasm is the main arena of cellular activities in eukaryotes only.
- (D) Activities of an organism are the outcome of sum total of activities and interactions of its constituent cells.

	A	B	C	D
(1)	T	F	F	F
(2)	F	T	F	T
(3)	T	F	F	T
(4)	F	F	T	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

121. Identify the correct ones w.r.t. centriole.

- a. It has 9 + 2 arrangement of microtubules.
- b. There are 9 peripheral triplets in it.
- c. The centre of the centriole possess a rod shaped mass of tubulin known as massule.
- d. It has a cartwheel organisation.
- e. It is surrounded by pericentriolar satellites.

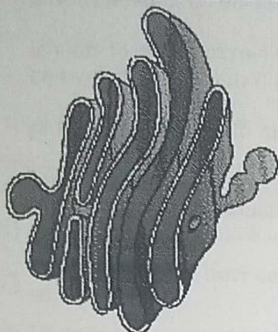
(1) a, b and d

(2) b, d and e

(3) b, c and d

(4) a, c and e

122. Identify the following figure and choose the option which has correct set of statements regarding the same.



- a. Disc-shaped sacs are concentrically arranged near the nucleus
 b. Has two entirely different but interconnected forming and maturing faces.
 c. Provides precursors of enzymes of lysosomes
 d. Site of formation of glycoproteins
 e. Packaging of materials only for intracellular targets
- (1) a, b, c & d only
 (2) a, b, d & e only
 (3) All except c & e
 (4) All except d & e
123. In castor, oil storing leucoplasts are called
- (1) Elaioplasts
 (2) Amyloplasts
 (3) Chromoplasts
 (4) Aleuroplasts
124. Carotenoids are
- (1) Fat soluble pigments
 (2) Water soluble pigments
 (3) Both fat and water soluble pigments
 (4) Found in leucoplast only
125. Select the **incorrect** statement for ribosomes.
- (1) They are granular structures
 (2) They are composed of RNA and proteins
 (3) They are not surrounded by any membrane
 (4) Eukaryotes have 70S cytoplasmic ribosomes
126. Which of the following structures is/are **not** found in Mycoplasma?
- (a) Organelle for aerobic respiration
 (b) Peptidoglycan in the plasma membrane
 (c) 50S and 30S subunits of ribosomes.
 (d) Organelle with cartwheel appearance
- Choose the **correct** option.
- (1) (d) only
 (2) All except (c)
 (3) (b) and (d) only
 (4) (a) and (b) only
127. Higher plants lack
- (1) Dictyosomes
 (2) Nucleus
 (3) Cell wall
 (4) Centriole
128. The component of endomembrane system that have ribosomes
- (1) Is major site for synthesis of lipids
 (2) Is involved in protein synthesis and secretion
 (3) Is double membrane bound
 (4) Was discovered by Camillo Golgi
129. Select the odd **one** w.r.t shapes of bacteria
- (1) *Bacillus* - Rod like
 (2) *Coccus* - Spherical
 (3) *Vibrio* - Spherical
 (4) *Spirillum* - Spiral
130. Ribosomes are composed of
- (1) rRNA and tRNA
 (2) mRNA and proteins
 (3) mRNA and rRNA
 (4) rRNA and proteins
131. Which of the following is **not correct** w.r.t chloroplast?
- (1) The space enclosed by the inner membrane of chloroplast is the stroma.
 (2) Stroma contains a large number of flattened non-membranous sacs called stroma lamellae.
 (3) Stroma contains enzymes required for the synthesis of carbohydrates.
 (4) Thylakoids are arranged in stacks called grana
132. A polyribosome is
- (1) Association of larger and smaller subunit of ribosome
 (2) Chain of ribosomes attached to ER
 (3) Inclusion bodies of prokaryotes
 (4) Several ribosomes attached to a single mRNA

133. The orange colour of carrot roots is due to which of the given plastids?

- (1) Amyloplast
- (2) Chloroplast
- (3) Chromoplast
- (4) Leucoplast

134. Read the following **Assertion (A)** and **Reason (R)** and select the **correct** option.

Assertion (A) : In prokaryotes, ribosomes are associated with plasma membrane of the cell.

Reason (R) : Only one type of ribosomes having 40S and 30S subunits are found in prokaryotes.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

ZOOLOGY

136. Which of the following cells are exclusively present in specialized connective tissue?

- (1) Fibroblasts
- (2) Macrophages
- (3) Chondrocytes
- (4) Mast cell

137. Characteristic of smooth muscle fibres are

- (1) Spindle shaped, unbranched, uninucleated, unstriated, voluntary
- (2) Spindle shaped, unbranched, uninucleated, unstriated, involuntary
- (3) Cylindrical, unbranched, striated, multinucleated, voluntary
- (4) Cylindrical, branched, striated, uninucleated, involuntary

138. A secondary metabolite among the following which is considered as a toxin is

- (1) Morphine
- (2) Abrin
- (3) Concanavalin A
- (4) Curcumin

135. A prokaryotic cell does not have

- (1) Ribosomes
- (2) Microtubules in flagella
- (3) Circular genomic DNA
- (4) Inclusion bodies

139. Choose the **correct** match w.r.t. types of tissues and their respective functions.

- (1) Compound epithelial tissue – Forms a diffusion boundary
- (2) Squamous epithelial tissue – Allows movement of mucus in a specific direction
- (3) Compound epithelial tissue – Conduction of dust particles in trachea
- (4) Columnar epithelial tissue – Secretion and absorption

140. Which of the following given features are true w.r.t. cardiac muscle fibres?

- (a) They are branched
 - (b) They do not get fatigued
 - (c) Nuclei are peripheral in position
 - (d) Non striated
 - (e) Cylindrical in shape
- Select the **correct** option.

(1) (a), (b) and (e)

(2) (a), (b), (c) and (e)

(3) (a), (b), (d) and (e)

(4) (a), (b) and (d)

141. Select the **incorrect** option w.r.t. neuroglial cells.

- (1) Play an important role in protection and support of neurons
- (2) Do not respond to stimulus
- (3) Make up less than 50% of the volume of neural tissue
- (4) Constituent of nervous tissue

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142. Complete the analogy by choosing the **correct** option.

Dense connective tissue : Ligament :: Loose connective tissue : _____

- (1) Areolar tissue
- (2) Dermis of skin
- (3) Blood
- (4) Bone

143. Select the **correct** option to complete the analogy.

Ciliated epithelium : Lining of bronchioles :: Simple columnar epithelium : _____

- (1) Lining of proximal convoluted tubules
- (2) Wall of blood vessels
- (3) Moist surface of buccal cavity
- (4) Lining of small intestine

144. Glucosamine and N-acetyl glucosamine are chemically modified

- (1) Nucleosides
- (2) Sugars
- (3) Lipids
- (4) Nucleotides

145. A group of similar cells along with intercellular substances performing a specific function is best described as

- (1) Cells
- (2) Tissue
- (3) Organ
- (4) Organism

146. During strenuous exercises, glucose is converted into _____ in the skeletal muscle of humans.

Fill in the blank with **correct** option.

- (1) Glycogen
- (2) Ethanol
- (3) Starch
- (4) Lactic acid

147. **Assertion** : Certain secondary metabolites are useful to "human welfare".

Reason : Coloured pigments, rubber, scents and antibiotics useful to humans are extracted from plants, fungal and microbial cells.

In the light of above statements, choose the **correct** option.

- (1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
- (2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are false statements

148. Read the following given statements and choose the **correct** option.

Statement A: All the carbon compounds that we obtain from living tissues can be called biomolecules.

Statement B: Living organisms also possess inorganic elements and compounds.

- (1) Both statements A and B are incorrect
- (2) Both statements A and B are correct
- (3) Only statement A is correct
- (4) Only statement B is correct

149. Read the given statements.

(a) Tendons and ligaments are examples of dense irregular connective tissue.

(b) Fibre secreting cells are absent in fluid connective tissue.

(c) Presence of lamellae is common to both limb bones and cartilage in the tip of nose.

(d) Most of the cartilages in vertebrates embryos are replaced by bones in adults.

Select the option with **correct** statements.

- (1) (a) and (b)
- (2) (b) and (d)
- (3) (c) and (d)
- (4) (a), (b) and (c)

150. Biomolecules present in living organisms are categorised into filtrate and retentate fractions during their biochemical analysis. Among these two categories which one contains cellulose?

- (1) Water soluble fraction
- (2) Both filtrate and retentate
- (3) Filtrate only
- (4) Retentate fraction

151. Which of the following is **not** a polymer of glucose?

- (1) Starch
- (2) Inulin
- (3) Glycogen
- (4) Cellulose

152. Glycosidic bonds are characteristic feature of which nitrogen lacking biomolecule?

- (1) Deoxyribonucleic acid
- (2) Insulin
- (3) Carboxypeptidase
- (4) Cellulose

153. The ash obtained from a tissue may contain how many of the following elements?

Sodium, Potassium, Calcium, Magnesium, Carbon, Oxygen, Hydrogen

- (1) 7
- (2) 6
- (3) 3
- (4) 4

154. Concanavalin A belongs to which category of secondary metabolites?

- (1) Drugs
- (2) Lectins
- (3) Pigments
- (4) Toxins

155. The most abundant chemical compound in living organisms is

- (1) Proteins
- (2) Nucleic acid
- (3) Water
- (4) Carbohydrates

156. Compound epithelium can be found in all of the following, except

- (1) Dry skin surface
- (2) Moist surface of mouth
- (3) Fallopian tube
- (4) Pharynx

157. Consider the following statements.

- (a) The intercellular material is solid and pliable, resisting compression
- (b) Cells are enclosed in small cavities called lacunae within the matrix secreted by them
- (c) The tissue interacts with smooth muscles to bring about body movements
- (d) It is the site of production of blood cells

The statements which are applicable to cartilage include

- (1) (a), (b) and (d)
- (2) (a), (c) and (d)
- (3) (c) and (d)
- (4) (a) and (b)

158. Exoskeleton of insects are made of a complex polysaccharide called

- (1) Inulin
- (2) Chitin
- (3) Starch
- (4) Cellulose

159. Match the following structures with their characteristic features and choose the correct option.

a. Tight junctions	(i)	Cement neighbouring cells together to form a sheet
b. Adhering junctions	(ii)	Transmit information through chemicals to another cell
c. Gap junctions	(iii)	Establish a barrier to prevent leakage of fluids or substances across the epithelial cells
d. Synaptic junctions	(iv)	Channels to facilitate communication between adjacent cells

- (1) a(iv), b(iii), c(i), d(ii)
- (2) a(ii), b(iv), c(i), d(iii)
- (3) a(iv), b(ii), c(i), d(iii)
- (4) a(iii), b(i), c(iv), d(ii)

160. Assertion (A) : Out of all the types of connective tissues, blood is the main circulating fluid that helps in the transport of various substances across the body.

Reason (R) : Blood is a loose connective tissue in which fibroblasts are absent.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) Both (A) and (R) are false
- (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

161. The gland that secretes saliva is categorised as

- (1) Endocrine gland
- (2) Unicellular gland
- (3) Multicellular gland
- (4) Ductless gland

162. Choose the odd one w.r.t. the products of glands released through tubes or ducts.

- (1) Earwax
- (2) Milk
- (3) Mucus
- (4) Insulin

163. Ligament connects

- (1) Muscle to muscle
- (2) Bone to muscle
- (3) Bone to skin
- (4) Bone to bone

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164. Choose the incorrect statement.

- (1) Some of the columnar cells bear cilia on their free surface and are called ciliated epithelium.
- (2) The cells in the columnar epithelium are tall and slender.
- (3) The nuclei of cells present in cuboidal epithelium are located at the base.
- (4) The epithelium made of more than one layer of cells provide protection against chemical stresses.

165. A teacher in the class once asked about the location of smooth muscles in the human body, to which the students gave various examples that are listed below:

- (a) Walls of blood vessels
- (b) Walls of heart
- (c) Stomach wall
- (d) Walls of intestine
- (e) Biceps

Choose the option that represents the correct set of examples given by the students. w.r.t. location of smooth muscles.

- (1) (a), (c) and (d)
- (2) (b), (c) and (d)
- (3) (a), (b) and (c)
- (4) (c), (d) and (e)

166. Which of the following provides blue black colour with I_2 solution?

- (1) Protein
- (2) DNA
- (3) Cellulose
- (4) Starch

167. Choose an incorrect statement w.r.t connective tissue.

- (1) In higher animals, connective tissues are most abundantly distributed in their body.
Collagen or elastin are the fibres of structural proteins
- (2) which provide strength, flexibility and elasticity to the tissues.
Ground substance or matrix consists of modified
- (3) polysaccharides secreted by the same cells which are responsible for the secretion of collagen or elastin.
Connective tissues are classified into two types i.e., soft
- (4) connective tissue that includes blood and lymph and specialized connective tissue that includes cartilage.

168. Long cylindrical fibres in muscles are composed of numerous fine fibrils called

- (1) Collagen fibres
- (2) Elastin fibres
- (3) Myofibrils
- (4) Fibroblasts

169. Read the following features.

- (a) Has a free surface which faces either body fluid or the outside environment.
 - (b) Cells present in it are compactly packed with little intercellular matrix.
 - (c) Provides a covering for some part of the body.
- All of the above are true for

- (1) Epithelial tissue
- (2) Muscular tissue
- (3) Nervous tissue
- (4) Connective tissue

170. The bond between two monosaccharides in a polysaccharide chain is

- (1) Peptide bond
- (2) Ester bond
- (3) Hydrogen bond
- (4) Glycosidic bond

171. The tissue that exerts the greatest control over the body's responsiveness to changing conditions have excitable cells called

- (1) Neuroglial cells
- (2) Neurons
- (3) Muscle cells
- (4) Mast cells

172. A six carbon sugar with aldehyde group is

- (1) Ribose
- (2) Deoxyribose
- (3) Glucose
- (4) Fructose

173. Lipids constitute _____ per cent of the total cellular mass. Select the correct option to fill the blank.

- (1) 10 – 15
- (2) 2
- (3) 5 – 7
- (4) 1

174. Which of the following acid is usually used to grind living tissue for the separation of micromolecules and macromolecules?

- (1) CH_3COOH
- (2) Cl_3CCOOH
- (3) C_2H_5COOH
- (4) C_3H_7COOH

175. The excess of nutrients which are not used immediately in the body are converted into fats and are mainly stored in _____.
Choose the option which fills the blank correctly.

- (1) Areolar connective tissue
- (2) Adipose connective tissue
- (3) Dense regular connective tissue
- (4) Dense irregular connective tissue

176. A type of specialised connective tissue which is rich in calcium salts and collagen fibres has a _____ ground substance. Select the correct option to fill in the blank.

- (1) Hard and pliable
- (2) Soft and non-pliable
- (3) Hard and non-pliable
- (4) Solid and pliable

177. Choose the **incorrect** match w.r.t. type of epithelium with its location and function.

	Epithelium	Location	Function
(1)	Ciliated	Oviducts	Passage of ovum through fallopian tube
(2)	Brush bordered cuboidal	PCT of renal tubule	Reabsorption of useful substances
(3)	Squamous	Wall of blood vessels	Exchange of materials between blood capillaries and tissue fluid
(4)	Columnar	Surface of skin	Highly resistant to mechanical abrasions

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

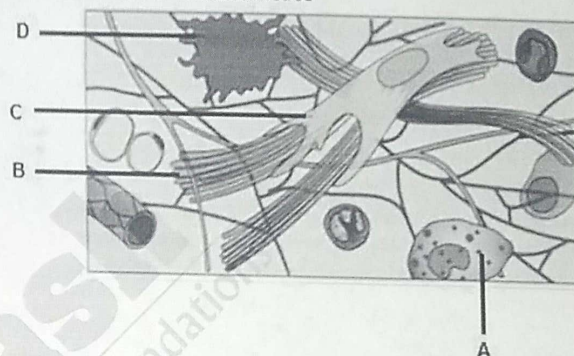
178. Biomolecule that is used for making paper from plant pulp is

- (1) Inulin
- (2) Cellulose
- (3) Glycogen
- (4) Chitin

179. Which of the following is a primary metabolite?

- (1) Carotenoids
- (2) Anthocyanins
- (3) Amino acids
- (4) Rubber

180. Following is the diagram of areolar connective tissue. Select the option which gives correct identification along with its functions/ characteristics



- (1) A – Mast cell, produces antibodies
- (2) B – Fibroblast, secretes matrix and fibres
- (3) C – Collagen, most abundant protein in animal world
- (4) D – Macrophage, ingests bacteria and foreign material